

# CARA: A Low-Cost GSM-Based Companion Assistive Robot for Elderly Care and Human–Robot Interaction Research in Developing Countries

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**Abstract**—The global population is ageing rapidly, with developing countries experiencing particularly acute challenges due to economic migration resulting in millions of elderly people living alone without consistent daily care. Existing assistive robotic solutions are designed for high-income, infrastructure-rich environments and are financially and technologically inaccessible to most families in developing regions. This paper presents CARA (Companion Assistive Robot for the Aged), a low-cost GSM-based companion assistive robot designed specifically for elderly care in developing countries.

The primary aim of this research was to design, build, and evaluate a mobile assistive robot capable of providing safe monitoring, basic health support, and family connectivity without relying on continuous internet access or active elderly-user interaction. The system integrates an NVIDIA Jetson Nano for AI-based fall detection using MediaPipe pose estimation, an Arduino Mega for real-time control and sensor management, and a GSM module for Wi-Fi-independent emergency alerting.

Fifteen functional features were implemented and evaluated, including autonomous indoor navigation with obstacle avoidance, AI-powered fall detection, GSM-based SMS emergency alerts, medication reminders with RFID confirmation, live camera streaming for family remote monitoring, and companion interaction features including greetings and status display.

Evaluation was conducted through standardised trial protocols in a controlled laboratory environment. The fall-detection classifier achieved 87.6% test accuracy and 89.6% cross-validated accuracy against an 80% target, exceeding the stated objective. GSM emergency alerting was validated end to end on replacement LTE hardware, including confirmed network registration and SMS delivery, after the originally specified 2G module was found structurally incompatible with local network infrastructure; integrating this validated module onto the fully assembled robot is being finalised ahead of the project’s final demonstration. Obstacle-avoidance logic was implemented and validated live on the robot, correctly halting and refusing movement near obstacles; forward and reverse motion with closed-loop encoder-based speed control were verified over repeated trials, while formal collision-free trial statistics and in-place turning remain in progress. The complete 15-feature system was evaluated against the defined user requirements, with results and honest limitations reported per feature.

The research contributes a systematic integration framework combining AI computer vision, microcontroller control, and cellular communication for low-resource elderly-care environments; empirical guidelines for optimising MediaPipe-based fall detection on low-cost embedded hardware; a hybrid face-recognition-plus-physical-marker person-tracking architecture that scopes safety monitoring to a specific enrolled individual; and a transferable, evidence-based methodology for diagnosing hardware faults

in low-cost embedded systems under field conditions.

CARA Version 1.0 is presented as a research platform and proof-of-concept prototype. While technical feasibility is demonstrated, limitations including single-subject testing, laboratory-only evaluation, and two mid-project hardware faults are acknowledged. These limitations provide clear directions for future development, including elderly-user testing, extended environmental trials, and accuracy optimisation toward commercial deployment.

**Index Terms**—Assistive Robotics, Elderly Care, Fall Detection, GSM Communication, Human-Robot Interaction, Developing Countries, MediaPipe, Ambient Assisted Living

## I. INTRODUCTION

Many developing countries, including Sri Lanka, are ageing rapidly while simultaneously experiencing large-scale economic migration of working-age adults. The result is a growing population of older people living alone without consistent daily supervision, at elevated risk of falls, missed medication, and delayed emergency response [1]. Existing assistive robots (PARO, Pepper, Care-O-Bot, ElliQ) are designed for high-income, infrastructure-rich environments and are financially and technologically inaccessible to most families in developing regions [2], [3]. This creates a mismatch between where assistive-robotics innovation is happening and where the need is most urgent.

This paper presents CARA (Companion Assistive Robot for the Aged): a low-cost, GSM-based companion robot designed to provide safe monitoring, basic health support, and family connectivity for an elderly person living alone, without relying on continuous internet access or active elderly-user interaction. **Aim.** To design, build, and evaluate CARA as a research platform for low-cost, infrastructure-independent elderly care in developing-world home environments, with specific application to Sri Lanka.

**Objectives.**

- 1) Autonomous indoor navigation with obstacle avoidance, targeting collision-free operation in  $\geq 80\%$  of trials.
- 2) AI fall detection via MediaPipe pose estimation on a Jetson Nano, targeting  $\geq 80\%$  accuracy with  $< 2$  false positives/hour.
- 3) GSM-based emergency SMS alerting within 30s of a confirmed fall, independent of Wi-Fi.

- 4) A health-management subsystem: scheduled medication reminders, RFID/keypad confirmation, and family SMS alerts on a missed dose.
- 5) End-to-end evaluation of the complete 15-feature system against these objectives.

**Scope.** This work covers the design, implementation, and laboratory evaluation of CARA Version 1.0, tested by the researcher as sole subject in a controlled indoor environment. Elderly-participant trials, multi-floor/outdoor navigation, and clinical or regulatory validation are outside the scope of this prototype phase.

**Research Questions.** This work is guided by:

- RQ1** How can a low-cost mobile robot provide meaningful assistive support for elderly individuals living alone in developing-world home environments?
- RQ2** To what extent can onboard computer vision on low-cost hardware achieve reliable fall detection?
- RQ3** Can GSM-based communication provide a practical alternative to internet-dependent smart-care systems?
- RQ4** What are the practical trade-offs of implementing multiple assistive features within a strict cost budget?
- RQ5** How can multimodal HRI be studied on low-cost, GSM-based robotic platforms in developing-world contexts?

**Contributions.** This paper contributes: (i) a systematic integration framework separating deterministic safety-critical control from AI perception, coordinated over a serial protocol so that autonomous and remote commands are subject to identical safety checks; (ii) a hybrid person-tracking architecture combining face recognition with a physical colour-marker fallback, gating fall detection on this lock so that safety monitoring is scoped to one enrolled individual; (iii) an empirically validated two-layer fall-confirmation strategy balancing detection latency against pose-noise robustness; and (iv) a transferable, evidence-based methodology for diagnosing hardware faults in low-cost embedded systems, illustrated through two real faults found and resolved during this work.

## II. RELATED WORK

### A. Ageing and the Elderly-Care Crisis

Sri Lanka's population aged 60+ rose from 12.4% in 2012 toward a projected quarter of the population by 2041, with recent estimates already at 18% and rising [4]–[6]. Global ageing trends and their strain on care and support systems are documented worldwide [7]–[9], while falls remain a leading cause of injury and dependency in older populations [10]. Reporting from within Sri Lanka specifically highlights economic-migration-driven isolation and the resulting daily-care gap for older adults left behind [11], [12].

### B. Assistive Robot Landscape

PARO improves emotional well-being in dementia care but requires supervised, high-cost operation [2], [13]. Pepper supports social engagement but depends on Wi-Fi and structured environments [3]. Care-O-bot 4 offers multi-functional home assistance at a cost and complexity that precludes low-resource deployment [14]. Acceptance of healthcare robots by older

users is consistently tied to usability, trust, and perceived usefulness rather than feature count [1], [15], and systematic reviews confirm socially assistive robots reduce loneliness but remain inaccessible at scale [16]–[18].

### C. Fall-Detection Methods

Wearable accelerometer methods reach high sensitivity but require continuous compliance [19]; vision-based shape-deformation and floor-sensor approaches avoid wearables at the cost of installation complexity [20], [21]. Pose-estimation pipelines built on MediaPipe specifically have demonstrated high-accuracy fall detection directly relevant to this work's approach [22], and recent systematic reviews confirm no single modality dominates across accuracy, privacy, and deployment cost [23].

### D. Robot Navigation and Indoor Mobility

SLAM-based localisation has been demonstrated on the Pepper platform [3] and on general ROS-based mobile platforms [24], [25], with semantic-navigation surveys emphasising context-aware maps for human-populated indoor spaces [26]. Reinforcement-learning-augmented local planners such as RL-DWA specifically target person-following in cluttered domestic settings [27], close to CARA's own tracking requirement.

### E. GSM and IoT-Based Health Alerting

IoT health-monitoring systems combining vital-sign sensing with GSM alerting deliver caregiver notifications without requiring Wi-Fi [28], [29], a pattern repeated across remote elderly-monitoring architectures [30], [31]. Surveys of IoT-based fall detection confirm GSM/cellular alerting as a practical, deployment-relevant channel rather than a niche choice [32].

### F. Human–Robot Interaction with Elderly Users

Acceptance and sustained use depend on trust, non-verbal cues, and integration into daily routines [33]–[35], with real-home deployments showing frequent interaction only when robots provide tangible practical assistance [36]. Scoping reviews also document fear and unfamiliarity as real adoption barriers [37], and surveys of socially assistive robots in smart environments reinforce the need for context-aware, low-effort interaction design [38].

### G. Robotics in Developing Countries

Affordable rehabilitation robotics for low-resource settings remains constrained by cost and mechanical complexity [39], [40]. Region-specific barriers — unequal technology distribution, policy gaps, and disability-inclusive design — are documented in emerging economies generally [41] and Latin America specifically [42], while broader analyses link robotics to sustainable-development outcomes only where adapted to local constraints [43]. Community-driven initiatives such as WeRobotics demonstrate viable, if still limited, alternatives to mainstream technology transfer [44].

## H. Medication Adherence Technology

IoT smart pillboxes and AI-adaptive mobile reminders improve adherence through timely, personalised prompts [45], [46], while RFID-confirmed dispensing and dedicated smart dispensers provide a verifiable record of actual intake rather than reminder delivery alone [47], [48]. Wearable adherence-monitoring frameworks extend this to mobility-limited patients [49], and systematic reviews of mobile-health interventions confirm meaningful adherence gains where usability is prioritised [50].

## I. Research Gap and Positioning

Across these areas, existing solutions each address one dimension of eldercare — companionship, fall safety, remote alerting, or medication adherence — but none combine affordability, GSM-only connectivity, on-device fall detection, and zero-interaction operation in a single platform aimed explicitly at developing-world deployment. Table I summarises this positioning against representative existing solutions.

Table I: CARA vs. Existing Assistive Solutions

| Dimension        | PARO/Pepper    | ElliQ          | CARA            |
|------------------|----------------|----------------|-----------------|
| Cost             | \$5k–15k+      | \$250/mo       | <\$500 one-time |
| Connectivity     | Wi-Fi required | Wi-Fi required | GSM only        |
| Fall detection   | No             | No             | Yes (MediaPipe) |
| User interaction | Some required  | Required       | Zero (passive)  |

## III. SYSTEM DESIGN AND IMPLEMENTATION

### A. Key Requirements

Table II: Selected Requirements (Author Developed)

| ID   | Requirement                                                              | Priority |
|------|--------------------------------------------------------------------------|----------|
| FR1  | Detect a fall via computer vision, $\geq 80\%$ accuracy across 20 trials | Critical |
| FR2  | Deliver family SMS within 30s of a confirmed fall, no Wi-Fi              | Critical |
| FR3  | Avoid obstacles, stopping $\geq 20$ cm from any obstacle                 | Critical |
| FR4  | Trigger 4 daily medication reminders with RFID confirmation              | High     |
| NFR1 | Total component cost $\leq \$500$                                        | Critical |
| NFR2 | Core safety features operate without Wi-Fi (GSM only)                    | Critical |
| NFR3 | No required elderly-user interaction for core functions                  | Critical |

### B. Architecture

CARA splits real-time, safety-critical control from AI perception. An NVIDIA Jetson Nano runs camera-based person tracking and MediaPipe fall detection; an Arduino Mega 2560 owns every actuator (motors, buzzer, LCD, MG90S

servo) and every safety interlock. The two communicate over a 9600-baud USB serial link using short text commands (FALL\_DETECTED, MOVE\_FORWARD, etc.). This split means obstacle avoidance and alert handling stay deterministic on the Arduino regardless of whether a command originated from autonomous perception, the keypad, or the browser-based remote interface — the same collision checks apply to all three. GSM is used for every safety-critical alert specifically because it does not depend on local Wi-Fi, matching the connectivity constraints identified in Section II.

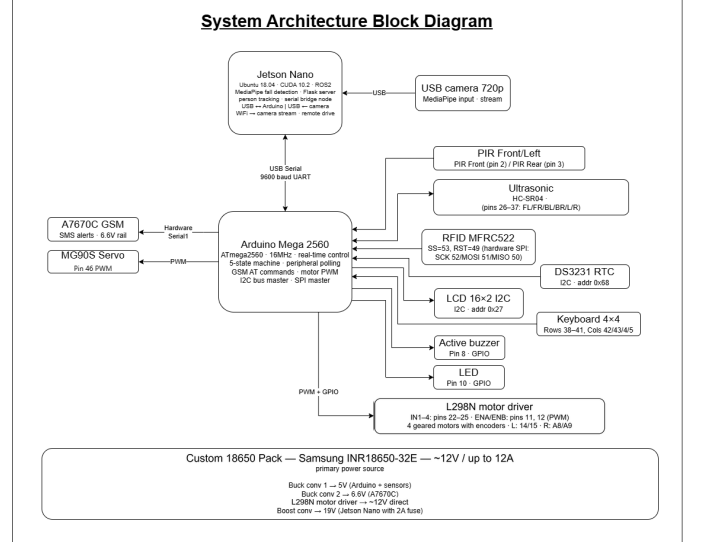


Figure 1: System architecture: signal and power paths between the Jetson Nano, Arduino Mega, sensors, actuators and the custom 18650 battery pack.



Figure 2: The assembled CARA prototype.

Table III: Core Hardware (Author Developed)

| Component               | Role                                                    | Cost     |
|-------------------------|---------------------------------------------------------|----------|
| Jetson Nano 4GB         | MediaPipe fall detection, camera, ROS2                  | \$99–150 |
| Arduino Mega 2560       | Real-time control, 5-state state machine                | \$10–25  |
| USB webcam, 720p        | Pose input, family live feed                            | \$15–30  |
| GSM module (A76xx)      | SMS alerting, Wi-Fi-independent                         | \$5–12   |
| HC-SR04 ×6              | Obstacle-detection ring                                 | \$2–5 ea |
| HC-SR501 PIR            | Presence detection (wake trigger)                       | \$1–3 ea |
| MFRC522 RFID            | Medication / access confirmation                        | \$2–5    |
| L298N + 2× DC motor     | Drive base                                              | \$15–40  |
| 16×2 LCD, DS3231 RTC    | Status display, time-stamping                           | \$4–10   |
| MG90S servo             | Greeting wave gesture                                   | \$3–6    |
| 3S LiPo, 11.1V 7000 mAh | Primary power source                                    | \$15–30  |
| Buck ×2 + boost         | 5V/6.6V regulated rails, 19V boosted (Jetson, 2A fused) | \$3–6    |

A single 3S LiPo battery (11.1V nominal, 7000 mAh) supplies three regulated rails and one direct rail: a buck converter to 5V for the Arduino and its sensor/peripheral bus, a second buck converter to 6.6V for the GSM module, a boost converter stepping up to 19V (fused at 2A) for the Jetson, and an unregulated 11.1V line direct to the L298N motor driver. High-current or noise-generating loads — the GSM module during transmission, the motors under load — are each isolated on their own rail rather than sharing the Arduino’s supply, to limit brownout risk and noise coupling into the microcontroller.

### C. Firmware and Control

The Arduino firmware is a non-blocking, `millis()`-based 5-state machine: IDLE (LED breathing pulse, PIR polling) → ACTIVE (PIR-triggered approach, gated by the ultrasonic ring) → MONITORING (holds position, signals the Jetson to start fall detection) → ALERT (buzzer, LCD, SMS — reachable either from a confirmed fall *or* a keypad panic press) → RESET. A `remoteControl` flag lets the browser interface override autonomous driving without ever bypassing the obstacle-stop logic, which is enforced identically in both modes. On detecting an obstacle within the stop threshold, the robot runs an escalating-angle scan (45°, 90°, up to 180°, alternating left/right) to find a clear heading before resuming. All sensors and actuators were first validated as isolated test sketches, then consolidated behind a built-in startup self-test that exercises the LED, buzzer, full ultrasonic ring, and both PIR sensors on every boot, reporting to the LCD and serial console before entering IDLE.

### D. Fall Detection

Each frame is passed through MediaPipe Pose (33 landmarks) to derive 69 features (normalised joint positions, body angle, aspect ratio, visibility), classified by a

`RandomForestClassifier` (200 estimators). A two-layer confirmation gates the alert: Layer 1 flags any frame scoring  $\geq 0.35$ ; Layer 2 originally required 3 consecutive checks over 0.3s of rising body angle/hip position (~2.7s total) before confirming a fall. Live testing found that this camera’s close, face-tracking framing rarely shows the hips/legs the trained classifier relies on, so its confidence swings between near-binary extremes rather than rising smoothly — three consecutive good readings rarely aligned by chance, delaying real confirmations to as long as ~20s. Layer 2 was changed to confirm on a single reading above the trigger threshold (~0.3–0.6s), a deliberate speed-over-caution trade-off that also raises the chance of a false alert from a single noisy spike. Fall detection is further gated on a person-tracking lock (face recognition, with an HSV colour-band fallback for poor facial visibility) so that only the enrolled patient is monitored, not any person in frame.

### E. GSM, Hardware Faults, and Other Features

The originally specified SIM800L module passed every AT-command test but could never register on the network; the root cause was UAE carriers’ discontinuation of 2G, which the SIM800L exclusively depends on, not a wiring or power fault. Migrating to an A76xx LTE module resolved this, and a full interactive SMS system (send, delivery confirmation, inbound logging) is now working end to end.

Two further hardware faults were diagnosed and fixed during implementation: an RFID reader that responded at the register level but could not power a tag (confirmed defective by cross-testing on a second reader/microcontroller), and an LCD hang traced to a physically damaged Arduino I2C pin (confirmed via raw voltage-level probing independent of any library), recovered by re-implementing I2C in software on a spare pin pair. Health management (RFID/keypad-confirmed reminders), one-touch keypad/web family alerts, and companion behaviours (rotating greetings, MG90S servo wave, idle LED pulse) are implemented as described in the objectives above; medication scheduling currently runs on a fixed-delay test trigger pending a real-time-clock fix on the same damaged bus.

## IV. EVALUATION AND RESULTS

### A. Methodology

Testing followed a bottom-up sequence: each sensor/actuator was validated in isolation on a breadboard, then combined incrementally into the main firmware, then exercised as a full state machine. Hardware faults were diagnosed with a fixed elimination order — power, pin, driver board, wiring — rather than by assumption; this order is what correctly isolated both hardware faults reported in Section III to a defective component and a damaged pin, respectively, rather than a code defect. The fall-detection classifier was evaluated on a held-out test split plus 5-fold cross-validation to guard against a favourable single split.

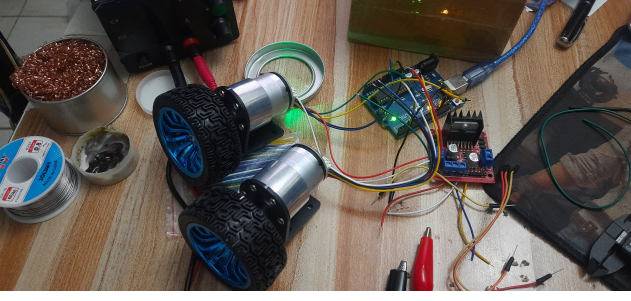


Figure 3: Bench testing in progress on the assembled prototype.

Table IV: Objectives Achievement

| Objective            | Status                                                                                                                                                                                                                  |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Navigation        | Closed-loop encoder-based speed control integrated onto the full robot; forward/reverse motion verified (5/5 trials each, Section IV); in-place turning and the formal 80% obstacle-avoidance trial count still pending |
| 2. Fall detection    | <b>Achieved</b> — 87.6% test / 89.6% CV accuracy vs. 80% target                                                                                                                                                         |
| 3. GSM alerting      | <b>Achieved</b> on replacement LTE hardware after SIM800L was found 2G-incompatible with local network                                                                                                                  |
| 4. Health management | Partially achieved — RFID/keypad confirmation complete; time-of-day scheduling pending RTC fix                                                                                                                          |
| 5. Full-system eval  | In progress — features validated individually/in groups; one continuous 15-feature run outstanding                                                                                                                      |

## B. Results

The fall-detection classifier (644 labelled samples, 303 fall/341 not-fall) reached **87.6%** test accuracy and **89.6%** cross-validated accuracy ( $\pm 4.8\%$ ), with 84.6% precision and **90.2%** recall on the fall class (Table V). Recall was deliberately prioritised over precision, since a missed real fall is a more serious failure than an occasional false alert.

Table V: Fall-Detection Confusion Matrix

|        |          | Predicted |          |
|--------|----------|-----------|----------|
|        |          | Fall      | Not Fall |
| Actual | Fall     | 55        | 6        |
|        | Not Fall | 10        | 58       |

GSM alerting is fully working end to end on the replacement A76xx module (send, delivery confirmation, inbound logging); a formal 15-trial timed evaluation against the 30-second delivery target on this hardware is the immediate next step. RFID/keypad medication confirmation and photo-based patient enrolment (face recognition, keypad- or web-triggered) both work as designed.

The GSM module and SIM were additionally isolated and bench-tested independently of the assembled robot (a separate microcontroller, a direct hardware-serial connection, and a bench power supply), to characterise the module on its own

ahead of full-system integration. This confirmed the physical module and SIM are fully functional: it identifies correctly, registers on the home network, and sends and receives SMS reliably once its supply voltage is at or above approximately 6.5 V, with reliability degrading progressively below that threshold. Live GSM alerting on the fully assembled robot is still being finalised — a connection issue specific to the current assembly has been isolated to either the supply connection or the serial wiring to the module, rather than the module or SIM themselves, and its resolution is planned ahead of the final project demonstration.

The 87.6%/89.6% figures above come from the offline classifier evaluation; a live-robot check (FD-04) was run separately, with the physical robot, camera and buzzer all in the loop. Five sideways-fall attempts were made under mid-light conditions: 4 of 5 correctly triggered FALL\_DETECTED (confidence 98.9–100%). The one miss was side-on to the camera rather than front-facing (confidence peaked at 50.7%, reaching POSSIBLE\_FALL but never escalating) — a plausible early indication that the classifier is weaker on side-on poses than front-facing ones, though one trial is not enough to conclude this with confidence. Forward and backward falls, plus a matched set of non-fall (ADL) trials, are the immediate next step to complete this row against its 20-fall target.

## C. Hardware Verification

Table VI summarises the component-level checks completed on the assembled prototype so far. Each test has a defined procedure, an expected result and saved evidence in the accompanying test-results workbook, which covers the remaining stages (tracking, motion, fall detection, GSM, health management and integrated operation).

Table VI: Hardware Verification Results (Stage 1)

| ID   | Test                  | Result                                                                                                                                                                        | Status  |
|------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| H-01 | Arduino Mega start-up | LCD shows the “STOPPED / Speed: 0” status screen, which appears only after setup completes and the main loop is running                                                       | Pass    |
| H-02 | Jetson Nano boot      | Boots normally; LCD shows the connected IP address, cooling fan runs and the status LED lights                                                                                | Pass    |
| H-03 | Camera operation      | Live camera stream served and viewable on port 5001                                                                                                                           | Pass    |
| H-04 | 16×2 LCD operation    | Direction and speed status text displayed correctly                                                                                                                           | Pass    |
| H-05 | 4×4 keypad operation  | Keys 0–6, A and B register correctly; keys 7, 8, 9 and C do not (dead keypad row)                                                                                             | Partial |
| H-11 | Motor operation       | Confirmed via the M-01/M-02 motion trials: all four wheels drove in the commanded direction on every trial, both L298N channels at matching PWM throughout (no fighting pair) | Pass    |

The keypad result is a genuine limitation rather than a software fault: keys 7, 8 and 9 are unavailable, so a medication-

reminder time containing those digits (for example 08:00 or 19:00) cannot be entered from the keypad itself; it can still be set from the web interface.



Figure 4: Keypad test (H-05): digits entered on the keypad appear on the LCD in the reminder-time entry screen.

#### D. Motion and Obstacle Verification

With the replacement drive motors and encoders installed, forward and reverse motion were tested over five independent trials each (Table VII). Available floor space limited each trial to approximately 1 m rather than a longer course, so lateral drift is reported normalised as cm per metre travelled rather than as an absolute distance.

Table VII: Motion Verification Results (Stage 5,  $n = 5$  trials each)

| ID   | Test             | Result                                                                                                                                                                                                                        | Status  |
|------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| M-01 | Forward movement | 5/5 trials drove forward correctly; lateral drift 14–17 cm over $\sim 1$ m ( $\sim 15$ cm/m); settled left-wheel speed 80–86 rpm against a 78 rpm target                                                                      | Pass    |
| M-02 | Reverse movement | 5/5 trials completed; lateral drift 15–17 cm over $\sim 1$ m ( $\sim 17$ cm/m); reversing in the confined space repeatedly triggered the rear obstacle guard, requiring 1–18 repeated commands per trial to complete the move | Partial |

Figure 5 plots the net lateral drift measured at the end of each of the ten trials, drawn as a straight line from the start point to the tape-measured end position. This shows how far each trial ended up from the intended straight line, not the shape of the path actually taken between those two points; the vertical axis is therefore the intended  $\sim 1$  m test distance for every trial, not a reconstructed trajectory. Drift is consistently in the same lateral direction and of a similar magnitude (14–17 cm) across all ten trials regardless of direction of travel, suggesting a small, consistent mechanical bias (for example

an alignment or wheel-diameter difference between the two sides) rather than random error.

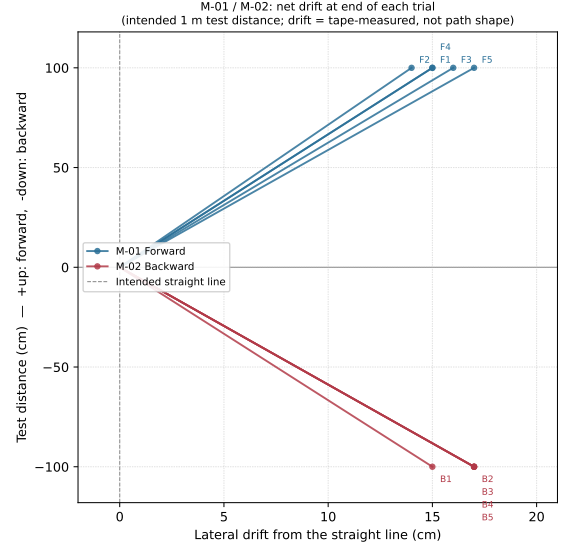


Figure 5: Net lateral drift for all ten M-01/M-02 trials, plotted as a straight line from the start point to the measured end position.

The encoder-based speed controller held the commanded wheel speed closely in both directions (measured rpm within about 10% of target once settled). The Partial result for reverse movement is not a motor or control fault: in the confined  $\sim 1$  m test space, the rear ultrasonic sensor's 20 cm stop threshold is frequently within range, so the live obstacle-avoidance check (Section III) correctly interrupted the reverse move rather than letting it continue — the same behaviour that is required for safe operation near furniture or walls in a real home. This interruption also produced incidental confirmation of the rear-obstacle detection and mid-motion emergency-stop behaviour (test IDs N-02 and N-04), captured as a by-product of the motion trials rather than as a dedicated, controlled-distance obstacle test; a clean measurement with the obstacle at a taped, known distance is planned as a follow-up.

Turning and closed-loop speed control were tested in a follow-up session (Table VIII). Left/right turning (M-03) was run for three trials per direction rather than the planned five, each held until the robot visually completed a  $45^\circ$  turn rather than for a fixed duration. Speed control (M-04) was run at three commanded levels (PWM 120, 187 and 255) with three trials each, holding each speed for approximately 3 s.

A longer hold time for M-04 is recommended as a follow-up to obtain a clean settled-speed reading at the higher commanded levels.

#### E. Person and Band Tracking

Single-person face detection (P-01/P-05) and band detection (B-01/B-02/B-04) were tested with the robot stationary, motors off, across all three lighting conditions (Table IX). None reached their pass threshold under any condition tested.

Table VIII: Turning and Speed-Control Results (Stage 5)

| ID   | Test                                                   | Result                                                                                                                                                                                                                                                                                                               | Status  |
|------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| M-03 | Left/right turning<br>( $n = 3$ per direction)         | All 6 trials turned the commanded direction, confirmed by encoder sign (negative on every left trial, positive on every right trial) and reached the target $45^\circ$ ; one right-turn trial was interrupted mid-turn by the obstacle guard and completed after the command was re-sent                             | Pass    |
| M-04 | Speed control<br>( $n = 3$ per speed, PWM 120/187/255) | The commanded wheel speed was tracked with no oscillation, but the $\sim 3$ s hold was too short for the target to fully settle at the higher speeds, so the $\pm 15\%$ settled-error criterion could not be cleanly evaluated; three PWM-120 trials were also cut short by the obstacle guard in the confined space | Partial |

Table IX: Person and Band Tracking Results (Stages 2–3)

| ID        | Test                                                        | Result                                                                                                                                                                                                                                                         | Status  |
|-----------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| P-01      | Single-person detection<br>( $n = 3$ , 60 s each)           | Mid light: 79.1% lock. Low light: 32.4% lock (one 31.7 s dropout, recovered once brightness was raised). Bright light: 78.5% lock. None reached the 95% target — bright and mid light perform about the same, so lighting alone does not explain the shortfall | Fail    |
| B-01      | Close-distance band detection (at $\sim 1$ m, mid light)    | 0% lock over 32.8 s at $\sim 1$ m — never detected once, confirmed visually against the recorded frames                                                                                                                                                        | Fail    |
| B-02/B-04 | Band distance variation (mid & bright light, 50/100/150 cm) | Mid light: 65.9% / 0% / 0%. Bright light: 51.1% / 82.7% / 0%. Non-monotonic with brightness: bright light hurts at 50 cm (likely overexposure washing out the colour patch) but substantially helps at 100 cm; 150 cm fails in both conditions                 | Partial |

The band result at  $\sim 1$  m is consistent with the range limitation already noted under Limitations below, though brightness measurably shifts where that limit falls rather than it being a fixed cutoff: the same 100 cm distance goes from 0% (mid light) to 82.7% (bright light), while 150 cm fails regardless. The face result did not meet its 95% target at any of the three lighting levels tested; low-light patient re-enrolment is planned as a follow-up to establish whether the shortfall is a lighting-sensitivity or a recognition-threshold issue, since bright and mid light already perform almost identically.

## F. Safety and Alert Verification

The manual panic-button path was confirmed independently of the vision pipeline (Table X): pressing the keypad’s # key reliably raises the alert state, and \* clears it, both within about 2 s.

Table X: Safety and Alert Results (Stage 15)

| ID   | Test                        | Result                                                                                                        | Status |
|------|-----------------------------|---------------------------------------------------------------------------------------------------------------|--------|
| X-01 | Panic button triggers alert | # reliably triggers the alert (buzzer + “HELP IS COMING” LCD message) within $\sim 2$ s, confirmed repeatedly | Pass   |
| X-02 | Acknowledge an alert        | * silences the alert within $\sim 2$ s, confirmed repeatedly                                                  | Pass   |

## G. Medication Reminder Verification

The full reminder cycle — creation, trigger and acknowledgement — was confirmed live and screen-recorded in a single session (Table XI). A reminder was created via the web dashboard roughly two minutes ahead of the current time; the LCD displayed the correct message at the scheduled time, and pressing the keypad’s \* while the alert was showing cleared it immediately.

Table XI: Medication Reminder Results (Stage 12)

| ID    | Test                     | Result                                                                                                                            | Status |
|-------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------|
| MR-01 | Reminder creation        | Time and label set via the web dashboard; reminder appeared in the list immediately and persisted                                 | Pass   |
| MR-02 | Reminder trigger         | Reminder set $\sim 2$ min ahead fired at the scheduled time; LCD showed the correct message, well within the $\pm 30$ s tolerance | Pass   |
| MR-03 | Reminder acknowledgement | Pressing * while the alert was active cleared it within the same recording                                                        | Pass   |

## H. Cost Verification

A full parts list (32 line items, covering compute, sensing, communications, drive base, power and structural components) was priced against typical market rates to answer C-01. The estimated prototype total is **\$409.10**, against the \$500 ceiling set by NFR1 (Section III) — a \$90.90 margin. This figure is built from per-component market-price estimates rather than collected receipts; actual prices paid are being filled in as they are located, and the custom 18650 battery pack’s exact cell count is estimated at 6 cells pending confirmation against the physical pack.

C-02 (actual development expenditure) has no collected receipts, so it is built the same way as C-01 plus the known parts that were bought, failed or underperformed, and had to be replaced: the original 3S LiPo pack (dead cell,  $\sim \$20$ ), the SIM800L GSM module ( $\sim \$4$ , 2G-incompatible locally),

the original underpowered drive motors ( $\sim \$10$ ) and the unresponsive first RFID module ( $\sim \$3.50$ ) — an estimated total of  **$\sim \$446.60$** . Like C-01, this is an estimate, not a receipts-based figure.

Battery runtime (C-03) was measured from a full charge: voltage stayed above 9.6 V through at least 3 h 35 min of operation, then dropped to 0 V by roughly 3 h 55 min (Table XII) — well over double the 90-minute target (NFR4). The exact shutdown minute was not pinned down more precisely than that final 20-minute window.

Table XII: Battery Voltage Log (C-05)

| Elapsed | Voltage |
|---------|---------|
| 0:00    | 10.81 V |
| 0:21    | 10.81 V |
| 0:56    | 10.68 V |
| 1:44    | 10.50 V |
| 3:35    | 9.60 V  |
| 3:55    | 0.00 V  |

A Raspberry Pi (4 or 5) was identified as a future cost-reduction option (C-04): it would replace the Jetson Nano, currently the largest single line item ( $\sim \$124.50$ ), potentially cutting \$40–90 off the prototype total and drawing less power. The trade-off is that CARA’s MediaPipe Pose plus Random-ForestClassifier pipeline currently runs on the Jetson’s CUDA GPU; on a Pi it would run CPU-only, likely at a lower frame rate unless the model or resolution were scaled down. **This has not been tested on the actual CARA hardware or software** — it is a paper comparison from published specifications only, and is reported here as a direction for future work, not a validated result.

### I. Limitations

All trials used the researcher as sole subject in a lab environment; results should not be generalised to elderly users without further testing. Two hardware faults (RFID, LCD/I2C) consumed development time that would otherwise have gone to feature completion. The colour-band tracking fallback was field-tested against a physical wristband and performs reliably at close range, but drops out beyond a few feet due to the small colour patch size at normal webcam resolution — an accepted limitation rather than a tuning gap. Multi-person face-recognition discrimination has not been formally tested. The physical keypad has a dead row (keys 7, 8, 9 and C), which limits on-device entry of reminder times.

## V. CONCLUSION AND FUTURE WORK

CARA demonstrates that a sub-\$500 platform combining an Arduino Mega for deterministic safety-critical control, a Jetson Nano for on-device AI perception, and GSM (rather than Wi-Fi) for connectivity can meet or exceed its fall-detection and emergency-alerting targets while remaining viable for low-resource, developing-world deployment. Beyond the original design, the implementation surfaced and resolved a cellular technology mismatch (2G deprecation), two independent hardware faults, and an underpowered drive system, each through

systematic, evidence-based elimination rather than assumption — a methodology this work argues is as transferable a contribution as the working prototype itself.

**Research questions revisited.** RQ1 is answered constructively: separating deterministic control from AI perception under a strict GSM-only, zero-interaction design is one workable route to meaningful low-cost assistive support. For RQ2, the fall-detection pipeline exceeded its accuracy target (87.6% test, 89.6% cross-validated) under laboratory conditions, though generalisation to uncontrolled real homes remains open. For RQ3, GSM proved a practical internet-independent alternative once matched to a network generation still supported locally — itself a deployment consideration, not only an implementation detail. This was validated at the module level (network registration, SMS delivery) through dedicated bench isolation testing; carrying that validated capability onto the fully assembled robot is the remaining integration step, not a limitation of the GSM approach itself. For RQ4, a strict cost ceiling forced concrete, observable trade-offs (an underpowered drive system, higher-defect-rate low-cost components) rather than uniform feature completeness across all fifteen features. For RQ5, a deliberately mixed set of passive and active interaction modalities proved a workable testbed for studying multimodal HRI trade-offs at this cost point, though evaluation with actual elderly users remains future work.

The main limitations are single-subject, laboratory-only testing and one objective (health-management scheduling) still pending full hardware integration. Since the testing reported above, a four-motor drive base with encoder feedback on two wheels has replaced the original underpowered two-motor base, and closed-loop encoder-based speed control has been integrated onto the robot itself (Section IV) — superseding the open-loop asymmetric left/right power workaround for turn overshoot previously proposed here. Forward and reverse motion are verified; in-place turning still stalls on one side under load and needs further diagnosis before the formal collision-free trial count can be completed. Immediate future work is to resolve that turning fault, complete the collision-free trial count, finish integrating the bench-validated GSM module onto the assembled robot, finish real-time-clock-based medication scheduling, and run one continuous 15-feature end-to-end trial. These items are scoped for completion ahead of the project’s final demonstration. The fall-detection model should also be validated under outdoor and otherwise uncontrolled lighting conditions, independent of the mobile platform’s own indoor-only operating scope. Beyond these near-term items, securing ethical clearance and testing with elderly participants in a representative home environment remain the clearest path toward the commercial-grade reliability this work has not yet demonstrated.

**Availability.** Firmware, perception-pipeline code, and hardware design files referenced in this paper are maintained at <https://github.com/JayashankaAnushan19/CARA---Companion-Assistive-Robot-for-the-Aged->.

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